ECONOMETRICS I

Lecture 7

Assumptions and diagnostics of the classical model

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Introduction

We have made a number of assumptions in this lecture (e.g., heteroskedasticity, normality, $E(u|X) = 0$, etc.). Key questions:

- What is no longer valid if a particular assumption is violated?
- How to detect if an assumption occurred?

Key insights:

- Map of assumption $\rightarrow$ result
- Residual analysis and other tests
- Introduction to the F-tests and applications
- Useful matrix formula:
  when $y \sim N(\mu, V)$, then $Ay + b \sim N(A\mu + b, AVA')$

Introduction

	Dependent variable: Birth weight		
	(1)	(2)	(3)
Cigarettes per day	−0.463***	−0.463***	−0.589***
	(0.092)	(0.093)	(0.111)
Household income	0.093***	0.091***	0.054
	(0.029)	(0.032)	(0.037)
Mother's education years		0.014	−0.438
		(0.258)	(0.320)
Father's education years			0.494*
			(0.283)
Constant	116.974***	116.835***	118.074***
	(1.049)	(3.138)	(3.500)
Observations	1,388	1,387	1,191
R^2	0.030	0.030	0.033
Adjusted R^2	0.028	0.028	0.030
Residual Std. Error	20.063 (df = 1385)	20.075 (df = 1383)	19.842 (df = 1186)
F Statistic	21.28*** (df = 2; 1385)	14.15*** (df = 3; 1383)	10.05*** (df = 4; 1186)
Note:			*p<0.1; **p<0.05; ***p<0.01

What is affected here if our assumptions are invalid?

Assumptions of the classical model

Assumptions of the classical model

Assumption	Means	Required for
1: $\text{rank}(X) = k$	No perfect multicollinearity	Existence of $\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y$
2: $y = X\beta + u$	Correct functional specification	Unbiasedness: $\mathbb{E}[\hat{\beta}\|X] = \beta$ (also 1 & 3 needed)
3: $\mathbb{E}[u\|X] = 0$	Exogenous zero-mean errors	Unbiasedness: $\mathbb{E}[\hat{\beta}\|X] = \beta$ (also 1 & 2 needed)
4: $\mathbb{V}[u\|X] = \sigma^2 I_n$	Homoskedastic & nonautocorrelated errors	Formula $\mathbb{V}(\hat{\beta}\|X) = \sigma^2(X'X)^{-1}$ (also 1, 2 & 3 needed)
5: $u\|X \sim N(0, \sigma^2 I_n)$	Normally-distributed errors	Validity of t-tests, confidence intervals, etc. (also 1 - 4 needed)

Assumption 1: No perfect multicollinearity

- **Mathematical statement**: $\text{rank}(X) = k$
- **Practical meaning**: No regressor can be expressed as exact linear combination of others
- **Derivation of OLS estimator**:

$$\min_{\hat{\beta}} \sum_{i=1}^{n} \hat{u}_i^2 = \hat{u}'\hat{u} = (y - X\hat{\beta})'(y - X\hat{\beta})$$

$$\text{FOC:} \quad -2X'(y - X\hat{\beta}) = 0$$

$$X'X\beta = X'y$$

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y \quad \text{(requires A1 for existence}$$

- **Why A1 is needed**:
  - If $\text{rank}(X) < k$, then $\text{rank}(X'X) < k$
  - $(X'X)$ is singular and non-invertible
  - No unique solution for $\hat{\beta}$

Assumption 2: Correct functional form

- **Mathematical statement**: $y = X\beta + u$ means

$$\underset{(n\times 1)}{y} = \hat{\beta}_1 \underset{(n\times 1)}{1} + \hat{\beta}_2 \underset{(n\times 1)}{x_2} + \hat{\beta}_3 \underset{(n\times 1)}{x_3} + \ldots + \hat{\beta}_k \underset{(n\times 1)}{x_k} + \underset{(n\times 1)}{\hat{u}}$$

- **Features in** :

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X\beta + u) \quad \text{(using A2)} \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'u \\ &= \beta + (X'X)^{-1}X'u\end{aligned}$$

- **Interpretation**:
  - Linearity in parameters (right previous transformations)
  - Violation becomes relevant in conjunction with $\mathbb{E}[u|X] = 0$ (see next slide)

Assumption 3: Exogenous zero-mean errors

- **Mathematical statement** $\mathbb{E}[u|X] = 0$ means

$$\mathbb{E}[\underset{n\times 1}{u}|X] = \begin{bmatrix} \mathbb{E}(u_1|X) \\ \mathbb{E}(u_2|X) \\ \vdots \\ \mathbb{E}(u_n|X) \end{bmatrix} = \underset{n\times 1}{0}$$

  - Mean independence: $\mathbb{E}[u_i|X] = 0$ for all i
  - Implies also: $\text{cov}(x_j, u) = 0$ for all regressors $j = 1, \ldots, k$
  - I.e., regressor **shouldn't predict** any error u_i
- **Derivation of unbiasedness**:

$$\begin{aligned} \hat{\beta} &= \beta + (X'X)^{-1}X'u \quad \text{(from A1 \& A2)} \\ \mathbb{E}[\hat{\beta}|X] &= \mathbb{E}[\beta + (X'X)^{-1}X'u|X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[u|X] \\ &= \beta + (X'X)^{-1}X' \cdot 0 \quad \text{(using A3)} \\ &= \beta \end{aligned}$$

Assumption 3: Exogenous zero-mean errors

- **Violations occur when**:
  - **Relevant omitted variable correlated with included regressors** (lecture 6)
  - Also (as we will see later):
    - Measurement error in regressors (we will see this later)
    - Simultaneity/reverse causality (we will see this later)
    - Sample selection bias (we will see this later)
- **Consequences**:
  - Bias: $\mathbb{E}[\hat{\beta}|X] \neq \beta$
  - Inconsistency: plim $\hat{\beta} \neq \beta$ (more on this later)

Assumption 4: Homoskedasticity and nonautocorr.

- **Mathematical statement**: $\mathbb{V}[u|X] = \sigma^2 I_n$ or $\mathbb{E}[uu'|X] = \sigma^2 I_n$ (under A3) means

$$V[u|X] = \mathbb{E}[(u - \mathbb{E}[u])(u - \mathbb{E}[u])'|X] = \mathbb{E}[uu'|X]$$

$$= \begin{bmatrix} E[u_1^2|X] & E[u_1u_2|X] & \cdots & E[u_1u_n|X] \\ E[u_2u_1|X] & E[u_2^2|X] & \cdots & E[u_2u_n|X] \\ \vdots & \vdots & \ddots & \vdots \\ E[u_nu_1|X] & E[u_nu_2|X] & \cdots & E[u_n^2|X] \end{bmatrix}$$

$$= \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_{n\times n}$$

- **Components of A4**:
  - Homoskedasticity: $\mathrm{var}(u_i|X) = \sigma^2$ for all i
  - No autocorrelation: $\mathrm{cov}(u_i, u_j|X) = 0$ for $i \neq j$

Assumption 4: Homoskedasticity and nonautocorr.

- **Used to obtain variance-covariance matrix**:

$$\hat{\beta} = \beta + (X'X)^{-1}X'u \quad \text{(from A1 \& A2)}$$

$$\begin{aligned}
\mathbb{V}(\hat{\beta}|X) &= \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}])(\hat{\beta} - \mathbb{E}[\hat{\beta}])'|X] \\
&= \mathbb{E}[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|X] \quad \text{(using A2, A3)} \\
&= \mathbb{E}[((X'X)^{-1}X'u)((X'X)^{-1}X'u)'|X] \\
&= \mathbb{E}[((X'X)^{-1}X'u)\underbrace{(u'X(X'X)^{-1})}_{(\text{since}(AB)'=B'A')}|X] \\
&= (X'X)^{-1}X'\mathbb{E}(uu'|X)X(X'X)^{-1} \\
&= (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} \quad \text{(using A4)} \\
&= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} \\
&= \sigma^2(X'X)^{-1}
\end{aligned}$$

- Thus, if $\mathbb{V}(u|X) \neq \sigma I$ then $\mathbb{V}(\hat{\beta}|X) \neq \sigma(X'X)^{-1}$

Gauss-Markov Theorem

Gauss-Markov Theorem : Under assumptions A1 through A4 (linearity, no perfect multicollinearity, exogeneity, and spherical errors), the Ordinary Least Squares estimator is the **Best Linear Unbiased Estimator** (BLUE).

- **BLUE Definition**:
  - **Linear**: The estimator is a linear function of the dependent variable y
  - **Best**: Among all linear unbiased estimators, it has the smallest variance
- **Note**: Normality (A5) is not required for Gauss-Markov

Assumption 5: Normally distributed errors

- **Mathematical statement** $u|X \sim N(0, \sigma^2 I)$ means

$$\begin{bmatrix} u_1|X \\ u_2|X \\ \vdots \\ u_n|X \end{bmatrix} \sim N(\underset{n\times 1}{0}, \underset{n\times n}{\sigma^2 I_n})$$

  - A4 included in this mathematical representation.
  - Independently of the values taken by the regressors, each error term is drawn from an independent normal distribution with mean 0 and variance σ^2.
- **A5 is required for statistical inference** (t-tests, confidence intervals, F-tests)

Assumption 5: Normally distributed errors

Recall that $\hat{t}_j = \frac{\hat{\beta}_j}{\widehat{\text{s.e.}}(\hat{\beta}_j)} \sim t_{n-k}$ bases on the normality of $\hat{\beta}_j$.

Use rule $\boxed{y \sim N(\mu, V) \implies Ay + b \sim N(A\mu + b, AVA')}$ to show **t-test and co. depend on A1 to A5**:

$$
\begin{aligned}
u|_X &\sim N(0, \sigma^2 I) && \text{(using A3-A5)} \\
y|_X = X\beta + u|_X &\sim N(X\beta, \sigma^2 I) && \text{(using A2)} \\
y &\sim N(X\beta, \sigma^2 I)
\end{aligned}
$$

Hence, given that $(X'X)^{-1}$ exists (using A1):

$$
\begin{aligned}
\hat{\beta} = (X'X)^{-1}X'y|_X &\sim N((X'X)^{-1}X'X\beta, \\
&\qquad (X'X)^{-1}X'\sigma^2 X(X'X)^{-1}) \\
\hat{\beta}|_X &\sim N(\beta, \sigma^2(X'X)^{-1}).
\end{aligned}
$$

Tests for Multiple Linear Restrictions

F-Test: Testing multiple restrictions

	Dependent variable: Birth weight		
	(1)	(2)	(3)
Cigarettes per day	-0.463^{***}	-0.463^{***}	-0.589^{***}
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F-Test: Testing multiple restrictions

- **t-test**: one restriction $\boxed{q = 1}$ (e.g. $\beta_j = 0$)
- **F-test**: multiple restrictions $\boxed{q > 1}$ (e.g. $\beta_3 = \beta_4 = 0$)
- **Null hypothesis**: $H_0 : R\beta = r$ where:
  - R is $q \times k$ restriction matrix
  - r is $q \times 1$ vector of restricted values
  - $q =$ number of restrictions
- **Alternative hypothesis**: At least one restriction is violated

F-Test: Testing multiple restrictions

Example: Consider $y = a + bx_2 + cx_3$.

Hypothesis	Representation as $H_0 : R\beta = r$	Test
$H_0 : b = 0$	$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$	t
$H_0 : c = 5$	$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 5$	t
$H_0 : a + b = 1$	$\begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 1$	t
$H_0 : a = b$	$\begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$	t
$H_0 : a = 1, b = 2$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$	F
$H_0 : a = b = c = 0$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$	F

Distribution Under Null Hypothesis

- **Starting from OLS distribution**:

$$\hat{\beta}|_X \sim N(\beta, \sigma^2 (X'X)^{-1}) \quad \text{(using A1-A5)}$$

- **Distribution of restrictions under H_0**:

$$(R\hat{\beta} - r) \sim N(0, \sigma^2 R(X'X)^{-1}R')$$

$$(\sigma^2 R(X'X)^{-1}R')^{-\frac{1}{2}}(R\hat{\beta} - r) \sim N(0, I_q)$$

- **Quadratic form gives chi-square**:

$$(R\hat{\beta} - r)'[\sigma^2 R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r) \sim \chi^2_q$$

Deriving the F-Statistic

- **Unknown** σ^2: We estimate using residuals:

$$\frac{\hat{u}'\hat{u}}{\sigma^2} \sim \chi^2_{n-k}$$

- **F-statistic as ratio**:

$$F = \frac{[(R\hat{\beta} - r)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r)]/q}{\hat{u}'\hat{u}/(n-k)}$$
$$\sim F_{q,n-k}$$

- **Components**:
  - Numerator: Measures violation of restrictions
  - Denominator: Estimates error variance
  - Degrees of freedom: q restrictions, $n-k$ error df

Special Case: Testing Multiple Coefficients = 0

- **Restricted vs. unrestricted models**:

$$F = \frac{(\hat{u}_r'\hat{u}_r - \hat{u}'\hat{u})/q}{\hat{u}'\hat{u}/(n-k)} \sim F_{q,n-k}$$

or

$$F = \frac{(R^2_{\text{full}} - R^2_{\text{restricted}})/q}{(1 - R^2_{\text{full}})/(n-k)} \sim F_{q,n-k}$$

- **Where**:
  - $\hat{u}_r$: residuals from restricted regression
  - $\hat{u}$: residuals from unrestricted regression
  - q: number of regressors restricted to zero
- **Interpretation**: Measures improvement in fit from additional regressors

Special Case: Testing Multiple Coefficients = 0

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Intuition Behind the F-Statistic

- **Key insight**: Restricted model always has worse fit
- **Numerator**: $\hat{u}_r'\hat{u}_r - \hat{u}'\hat{u}$ measures:
  - Increase in sum of squared residuals
  - Explanatory power gained from q additional regressors
- **Large F-value indicates**:
  - Significant improvement in model fit
  - At least one of the q coefficients is statistically significant
  - Joint significance of the restricted regressors
- **Small F-value indicates**:
  - Little improvement from additional regressors
  - Restricted coefficients jointly insignificant

Software F-Test: Overall Model Significance

- **Common software output**: Tests all slopes (except constant) $= 0$

$$H_0 : \beta_2 = \beta_3 = \cdots = \beta_k = 0$$

- **Example**: $F(2, 10) = 13.59437$ with p-value $= 0.001406$
  - $q = 2$ restrictions
  - $n - k = 10$ degrees of freedom
  - Highly significant: Model has explanatory power
- **Interpretation**: At least one slope coefficient is statistically different from zero

Relationship Between t and F Tests

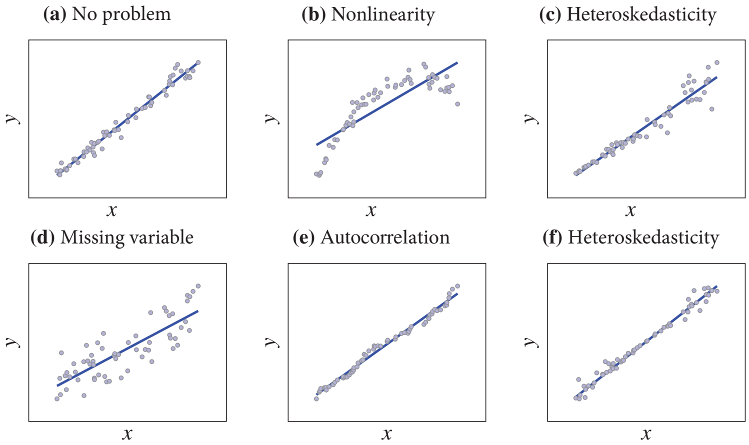
- **When** $q = 1$: F-test equivalent to two-tailed t-test

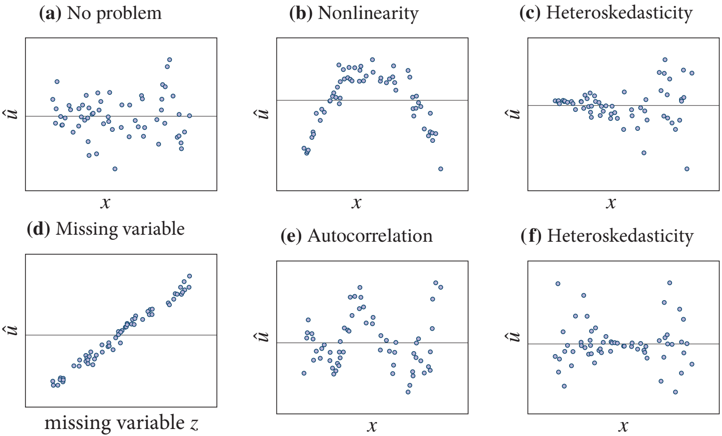
$$F_{1,n-k} = t^2_{n-k}$$

- **When** $q > 1$: F-test more appropriate than multiple t-tests
  - Controls overall Type I error rate
  - Tests joint significance
  - Accounts for correlation between estimators
- **All tests require A1-A5**:
  - Normality crucial for exact finite-sample distributions
  - Without A5, distributions are only asymptotically valid

Residual Analysis and Diagnostic Tests

- **Purpose**: Detect violations of classical assumptions, often using estimated residuals $\hat{u}$
- **Why residuals?**:
  - We observe $\hat{u} = y - X\hat{\beta}$, not the true errors u
  - Under A1-A5, $\hat{u}$ should behave like the true errors
  - Systematic patterns in $\hat{u}$ indicate model misspecification
- **Common diagnostic tests**:
  - F-tests for variable inclusion/exclusion
  - RESET test for functional form
  - Breusch-Pagan and White tests for heteroskedasticity
  - Normality tests (Jarque-Bera, etc.)


(a) No problem
(b) Nonlinearity
(c) Heteroskedasticity
(d) Missing variable
(e) Autocorrelation
(f) Heteroskedasticity
y
x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(e) Autocorrelation
(f) Heteroskedasticity
$\hat{u}$
x
missing variable z

F-Tests for Variable Inclusion

- **Testing relevance of variable subsets**:

$$H_0 : \beta_{k-q+1} = \beta_{k-q+2} = \cdots = \beta_k = 0$$

- **Implementation**:

$$F = \frac{(\hat{u}_r' \hat{u}_r - \hat{u}'\hat{u})/q}{\hat{u}'\hat{u}/(n-k)} \sim F_{q,n-k}$$

- **Where**:
  - $\hat{u}_r$: residuals from model excluding q variables
  - $\hat{u}$: residuals from full model with all variables
  - Large F-value suggests excluded variables are jointly significant
- **Example**: test $\gamma = 0$ in

$$y = X\hat{\beta} + Z\hat{\gamma} + \hat{u},$$

where Z is the matrix containing higher order polynomials of the original regression.

RESET Test for Functional Form

- **Purpose**: Detect incorrect functional form/misspecified nonlinearities
- **Procedure**:
  1. Estimate original model: $y = X\beta + u$, get $\hat{y}$ and residuals $\hat{u}$
  2. Augment model with powers of $\hat{y}$:
  $$y = X\beta + \gamma_1\hat{y}^2 + \gamma_2\hat{y}^3 + \cdots + \gamma_p\hat{y}^{p+1} + v$$
  3. Test $H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_p = 0$ using F-test
- **F-statistic**:
$$F = \frac{(SSR_r - SSR_u)/p}{SSR_u/(n-k-p)} \sim F_{p,n-k-p}$$
- **Interpretation**:
  - Significant F-value suggests functional form misspecification
  - Common choice: $p = 2$ (squares and cubes of fitted values)

RESET Test for Functional Form

- **Intuition behind RESET**:
  - If model is correctly specified, powers of $\hat{y}$ should have no explanatory power
  - $\hat{y}$ is linear function of X, so $\hat{y}^2, \hat{y}^3$ capture nonlinear patterns
  - Significant coefficients indicate omitted nonlinearities
- **Practical implementation**:

$$\hat{y} = X\hat{\beta}$$

$$F = \frac{(R^2_{\text{augmented}} - R^2_{\text{original}})/p}{(1 - R^2_{\text{augmented}})/(n-k-p)}$$

Breusch-Pagan Test for Heteroskedasticity

- **Testing A4**: $\mathrm{var}(u_i|X) = \sigma^2$ (homoskedasticity)
- **Procedure**:
  1. Estimate original model, get residuals $\hat{u}$
  2. Regress squared residuals on potential heteroskedasticity variables Z (usually just $Z = X$):

$$\hat{u}_i^2 = Z_i'\delta + v_i$$

  3. Test $H_0 : \delta_2 = \delta_3 = \cdots = \delta_m = 0$ using F-test

White Test for Heteroskedasticity

- **More general than Breusch-Pagan**: Doesn't require specifying form of heteroskedasticity
- **Procedure**:
  1. Estimate original model, get residuals $\hat{u}$
  2. Regress $\hat{u}^2$ on all regressors, their squares, and cross-products:

$$\hat{u}_i^2 = \delta_0 + \sum_{j=1}^{k} \delta_j x_{ji} + \sum_{j=1}^{k} \sum_{l \geq j}^{k} \delta_{jl} x_{ji} x_{li} + v_i$$

  3. Test joint significance of all coefficients except intercept using F-test
- **F-statistic**:

$$F = \frac{R^2_{\text{aux}}/(m)}{(1 - R^2_{\text{aux}})/(n - m - 1)} \sim F_{m,n-m-1}$$

where $m = 2k + \frac{k(k-1)}{2}$ for full White test

Normality Tests

- **Testing A5**: $u|X \sim N(0, \sigma^2 I)$
- **Jarque-Bera test**:

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2_2$$

where:

- $S = \frac{\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^3}{\left(\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2\right)^{3/2}}$ (skewness)
- $K = \frac{\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^4}{\left(\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2\right)^{2}}$ (kurtosis)

- **Null hypothesis**:
  - $H_0 : S = 0$ (symmetric) and $K = 3$ (mesokurtic)
  - Normal distribution has skewness = 0, kurtosis = 3

Normality Tests

- **Other normality tests**:
  - Shapiro-Wilk test (more powerful for small samples)
  - Kolmogorov-Smirnov test
  - QQ-plots (graphical method)
- **Consequences of non-normality**:
  - OLS still BLUE under A1-A4
  - CIs, t and F tests invalid in small sample
  - CIs, t and F tests asymptotically ok (more on this later)

Practical diagnostic strategy

1) **Most important: think about S2-S3** (e.g., omitted variables)
   1. Adj. R^2, residual plots, F-test, etc.: only limited help
   2. Rely on logic, good knowledge about the context of your study
2) **Less important but easy to do: check S4-S5**
   1. Graphical inspection
   2. Residual tests (e.g. White, Jarque-Bera, etc.)

- **General principle**: Let residuals guide model improvement, but deep questions (e.g. possible OVB) must be at the forefront.

Takeaways

Key takeaways

You should now know:

- Which assumptions are important for what
  - Feasibility (A1)
  - Bias (A2, A3)
  - Variance (A2-A4)
  - Statistical tests (A2-A4 + A5)
- The F-test as test to several coefficients at once
- Diagnostic tests
  - Visual recognition of residual plots
  - Understand basic ideas behind diagnostic tests
  - But remember: if A2 and A3 violated, then everything else wrong.

Appendix

χ^2 distribution

- **χ^2 Distribution:** If $Z_1, Z_2, ..., Z_n$ are independent standard normal variables, then

$$Z = \sum_{i=1}^{n} Z_i^2 \sim \chi_n^2 \qquad \text{and} \qquad \sqrt{Z} \sim \chi_n^2.$$

(Note: if k of the n normal variables are not independent, then $\sim \chi^2_{n-k}$)

t distribution

- t **Distribution:** Let Z_1 be a standard normal variable and Z_2 be a χ^2 variable with n degrees of freedom, that is $Z_1 \sim N(0,1)$ and $Z_2 \sim \chi^2_n$, then

$$t = \frac{Z_1}{\sqrt{Z_2/n}} = \frac{\sqrt{n}Z_1}{\sqrt{Z_2}} \sim t_n.$$

- Thus, in our regression

$$\frac{\hat{\beta}_j - \beta_{H0}}{\hat{s}(\beta_j)} \sim t_{n-k}$$

F distribution

- **F Distribution:** Let Z_1 and Z_2 be independent χ^2 variables with n_1 and n_2 degrees of freedom, respectively, then

$$F = \frac{Z_1/n_1}{Z_2/n_2} \sim F_{n_1,n_2}$$